

(12) **United States Patent**
Tao et al.

(10) **Patent No.:** **US 12,417,519 B2**
(45) **Date of Patent:** ***Sep. 16, 2025**

(54) **HDR TONE MAPPING METHODS WITH
SCENE ADAPTIVE PARAMETERIZED TONE
MAP FUNCTION**

(71) Applicant: **Samsung Electronics Co., Ltd.,**
Suwon-si (KR)
(72) Inventors: **Li Tao**, Mountain View, CA (US);
Yeong-Taeg Kim, Mountain View, CA
(US); **Chenguang Liu**, Mountain View,
CA (US); **Pradeep Nagesh**, Mountain
View, CA (US)
(73) Assignee: **Samsung Electronics Co., Ltd.,**
Suwon-si (KR)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.
This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/453,905**
(22) Filed: **Aug. 22, 2023**
(65) **Prior Publication Data**
US 2024/0046437 A1 Feb. 8, 2024

Related U.S. Application Data
(63) Continuation of application No. 17/512,603, filed on
Oct. 27, 2021, now Pat. No. 11,769,235, which is a
(Continued)
(51) **Int. Cl.**
G06T 5/00 (2024.01)
G06T 5/92 (2024.01)
H04N 19/00 (2014.01)

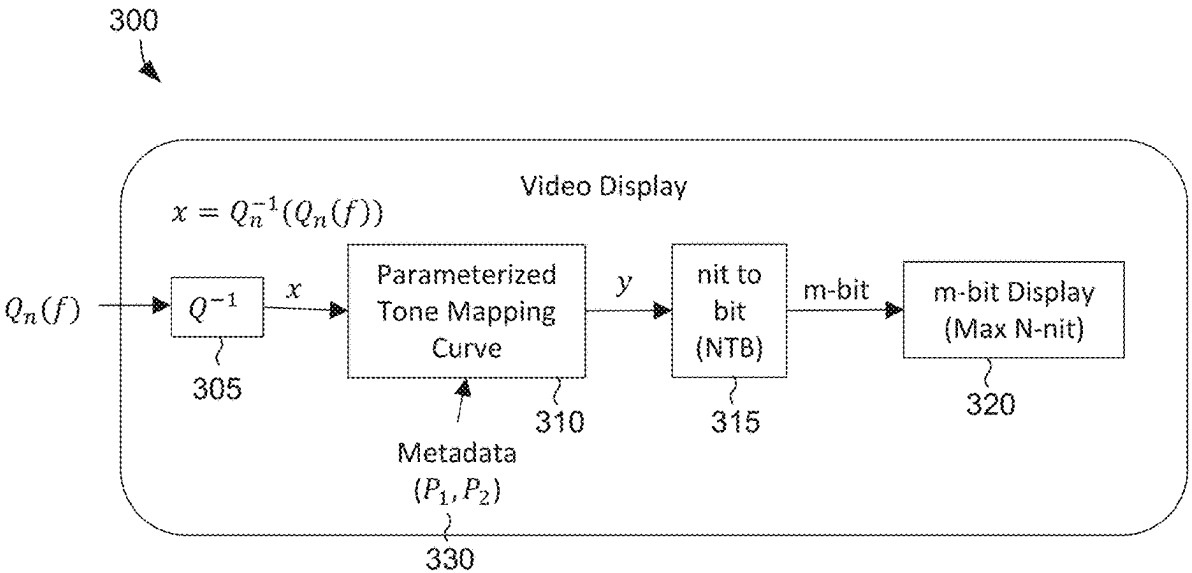
(52) **U.S. Cl.**
CPC **G06T 5/92** (2024.01); **H04N 19/00**
(2013.01); **G06T 2207/20208** (2013.01)
(58) **Field of Classification Search**
CPC .. **H04N 5/00**; **H04N 7/00**; **H04N 9/00**; **H04N**
19/00; **G06K 9/00**; **G06T 5/00**
See application file for complete search history.

(56) **References Cited**
U.S. PATENT DOCUMENTS
5,363,477 A 11/1994 Kuragano et al.
2004/0201610 A1 10/2004 Rosen et al.
(Continued)

OTHER PUBLICATIONS
Julian Faraway, Modelling three-dimensional trajectories by using
Bezier curves with application to hand motion, Apr. 2007, Univer-
sity of Bath, UK, Part 5, pp. 571-585 (Year: 2007).*
(Continued)
Primary Examiner — Maryam A Nasri

(57) **ABSTRACT**
A method includes receiving a quantized video having a first
maximum luminance level or brightness associated with a
mastering display with which the quantized video was
mastered. The method also includes de-quantizing the quan-
tized video in order to generate a de-quantized video. The
method further includes applying a scene-adaptive tone
mapping function to the de-quantized video in order to
generate a tone mapped video. In addition, the method
includes displaying the tone mapped video on a target
display. The tone mapped video has a second maximum
luminance level or brightness associated with the target
display, and the second maximum luminance level or bright-
ness is less than the first maximum luminance level or
brightness.

20 Claims, 8 Drawing Sheets



Related U.S. Application Data

continuation of application No. 14/986,563, filed on
Dec. 31, 2015, now Pat. No. 11,176,646.

- (60) Provisional application No. 62/184,667, filed on Jun.
25, 2015.

(56) **References Cited**

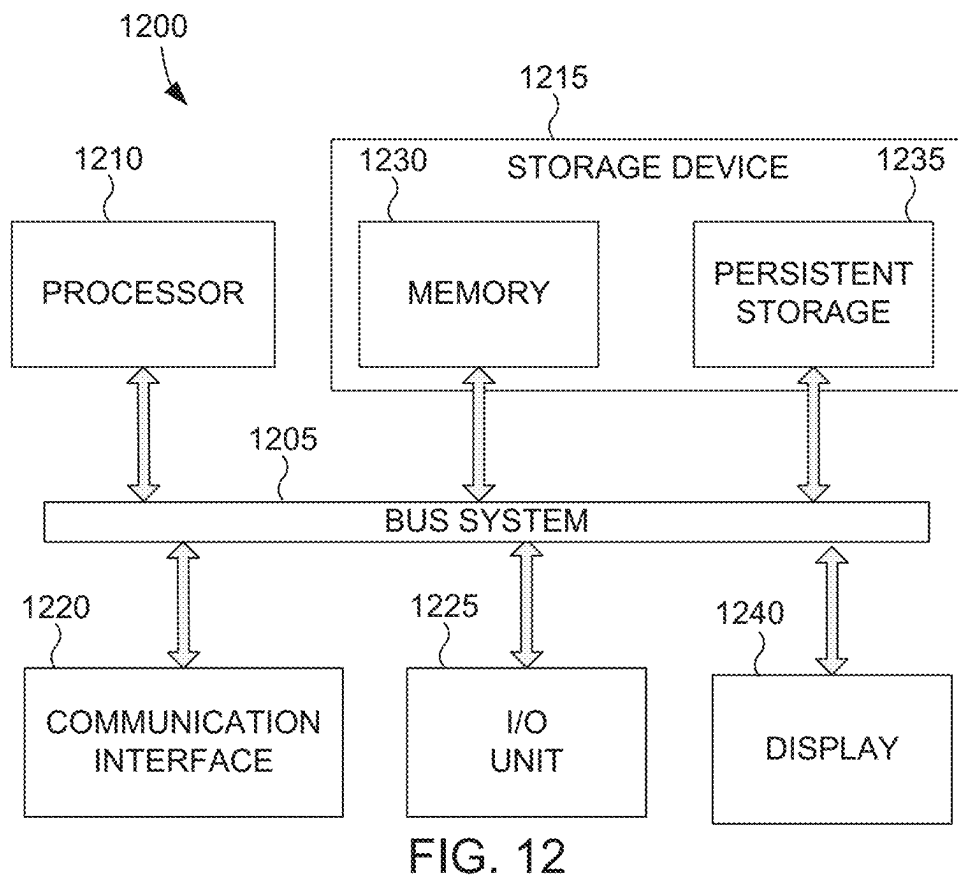
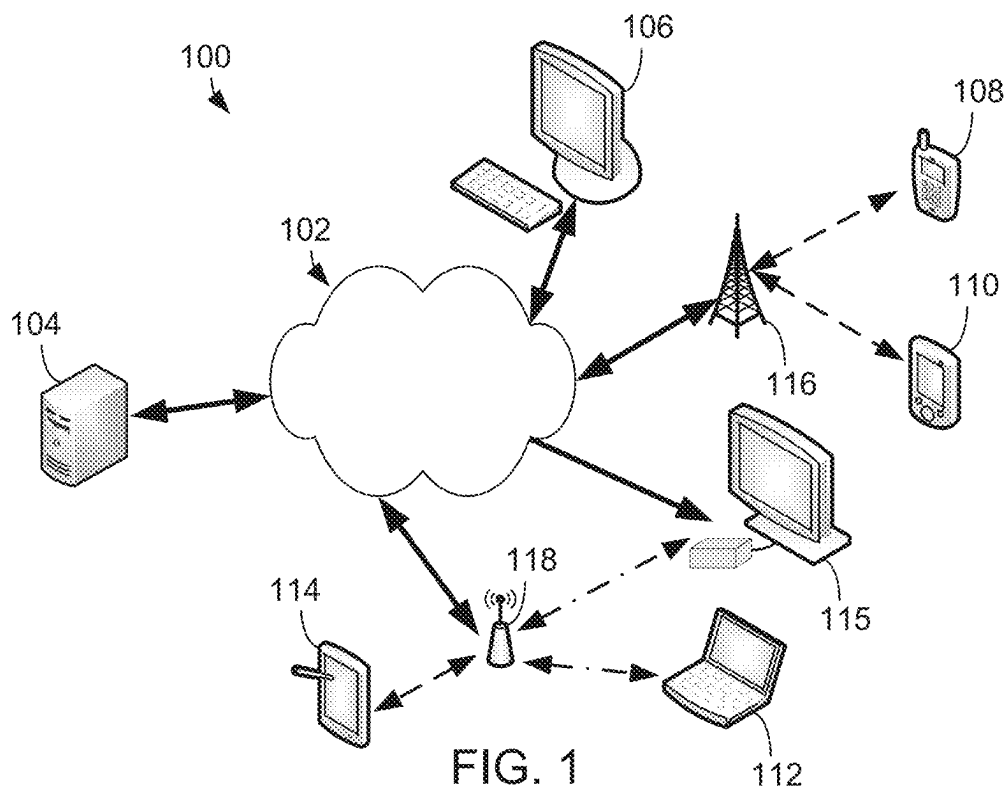
U.S. PATENT DOCUMENTS

2011/0050934	A1 *	3/2011	Mitsunaga	G06T 5/40 348/222.1
2012/0147953	A1 *	6/2012	El-Mahdy	G06T 5/92 375/240.03
2016/0189409	A1 *	6/2016	Aiba	G09G 5/10 345/629
2016/0366386	A1	12/2016	Douady-Pleven et al.	

OTHER PUBLICATIONS

Faraway et al., "Modelling three-dimensional trajectories by using
Bezier curves with application to hand motion," Journal of the
Royal Statistical Society Series C Applied Statistics, Nov. 2007, 17
pages.

* cited by examiner



200

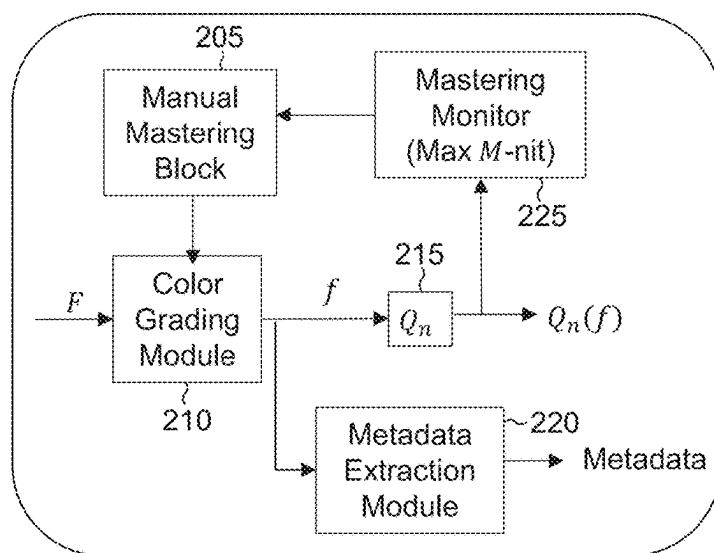


FIG. 2

300

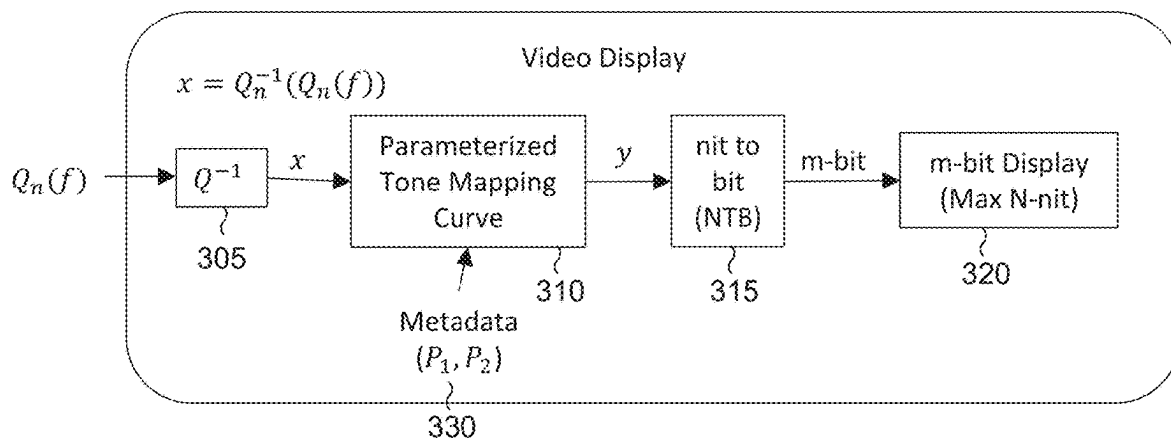


FIG. 3

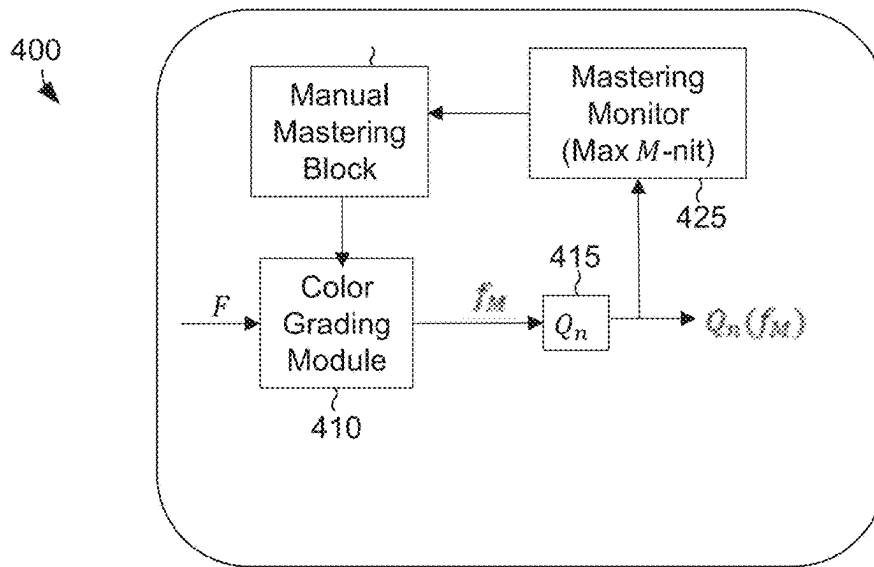


FIG. 4

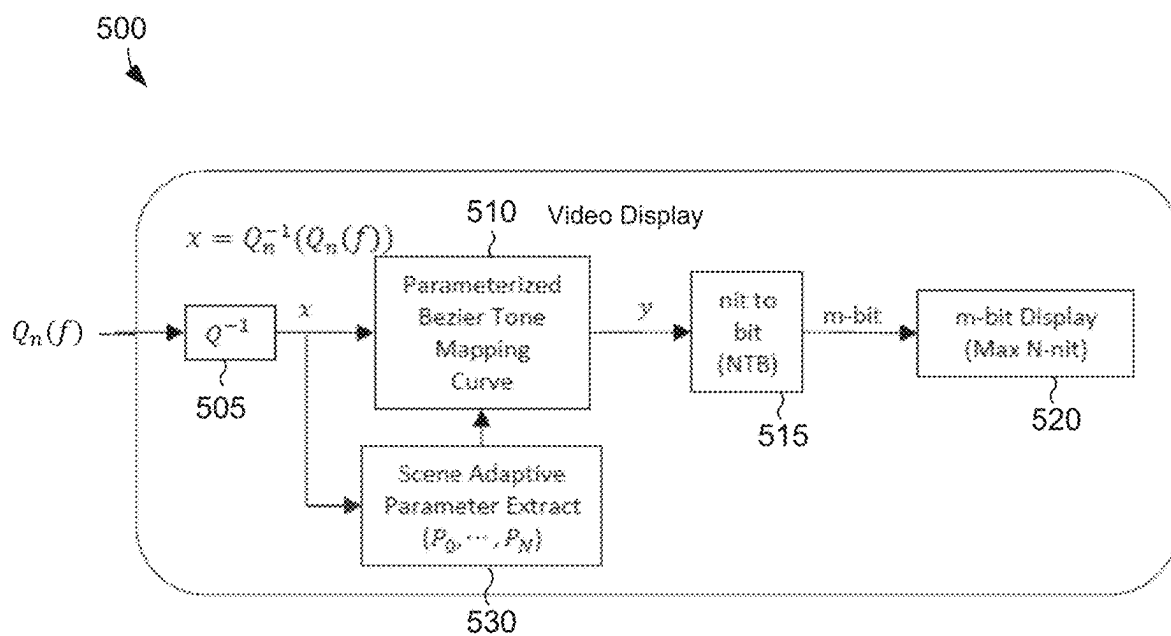


FIG. 5

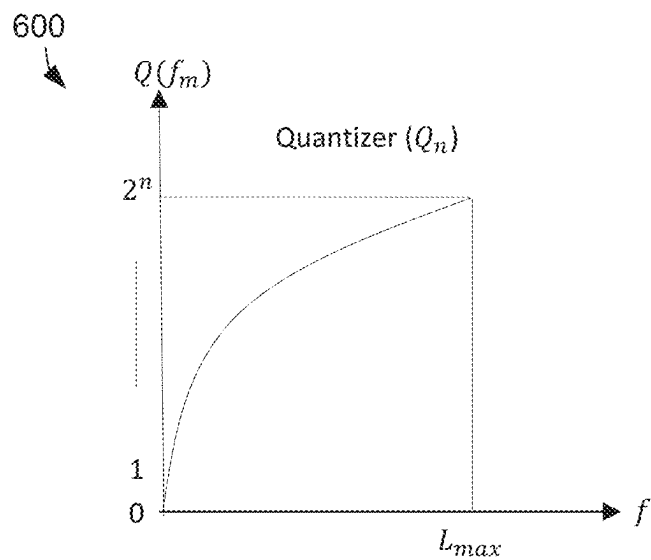


FIG. 6A

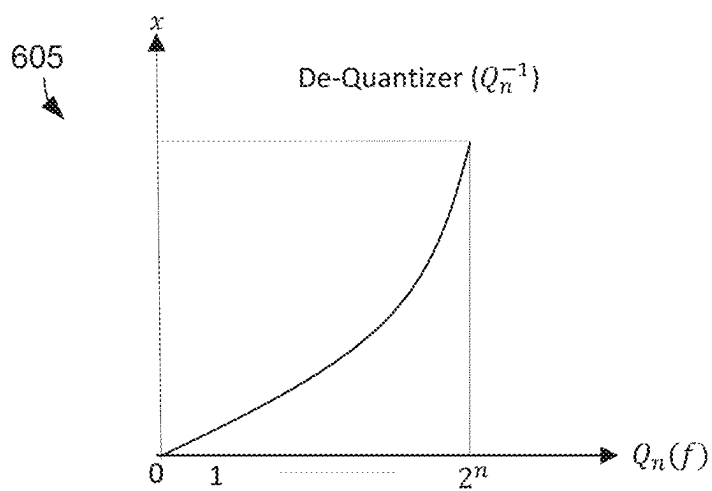


FIG. 6B

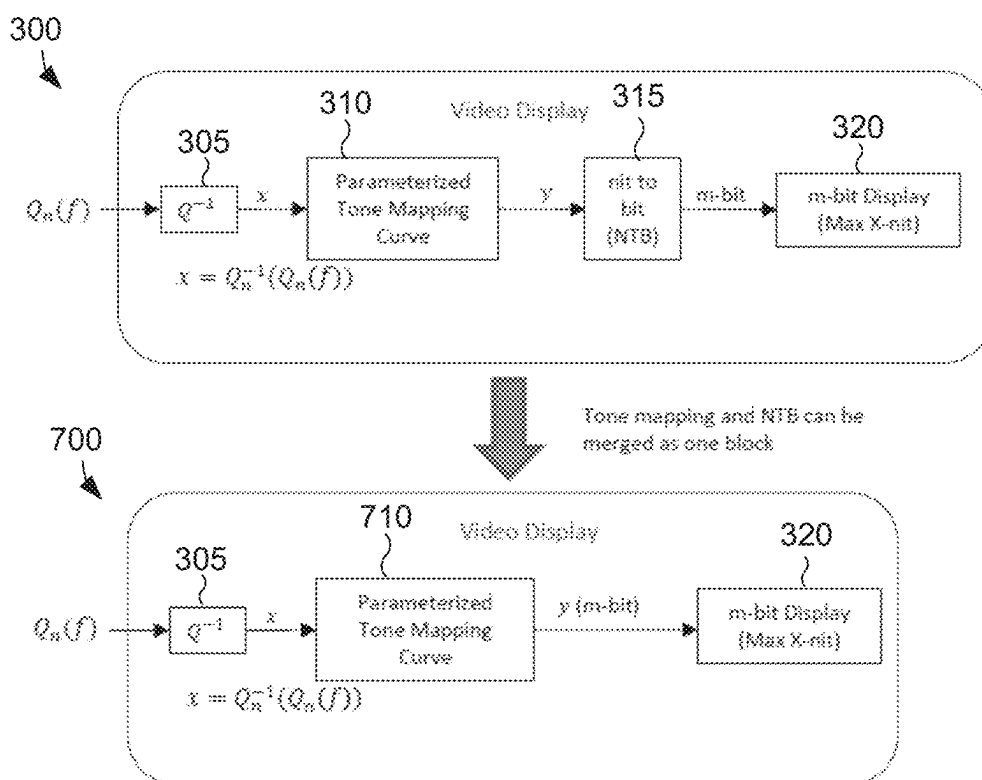


FIG. 7A

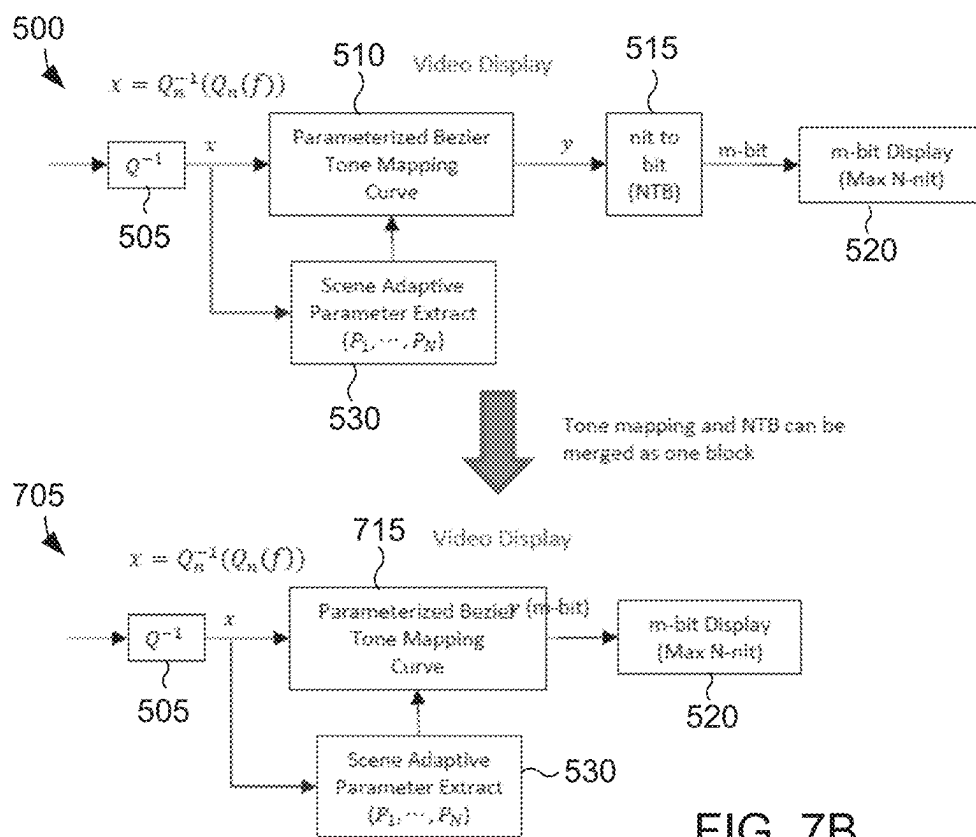


FIG. 7B

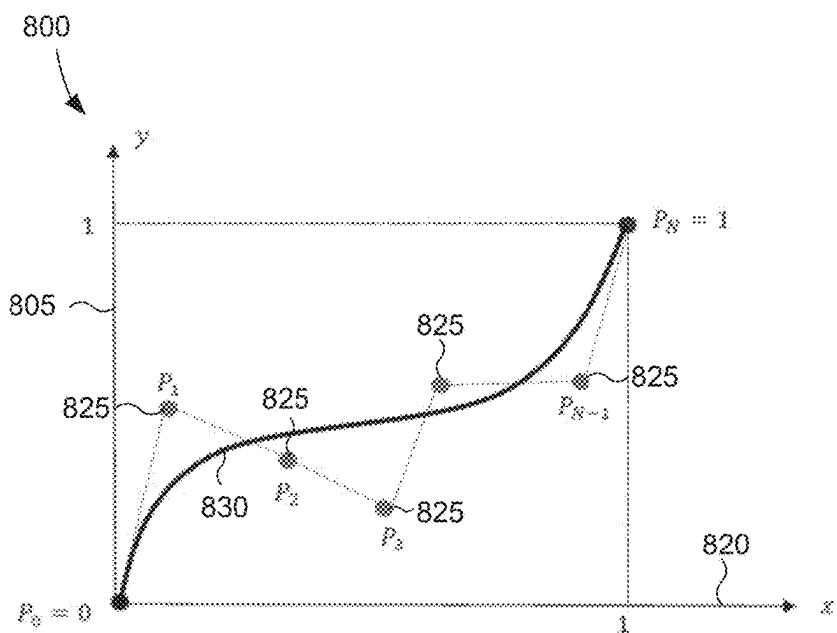


FIG. 8

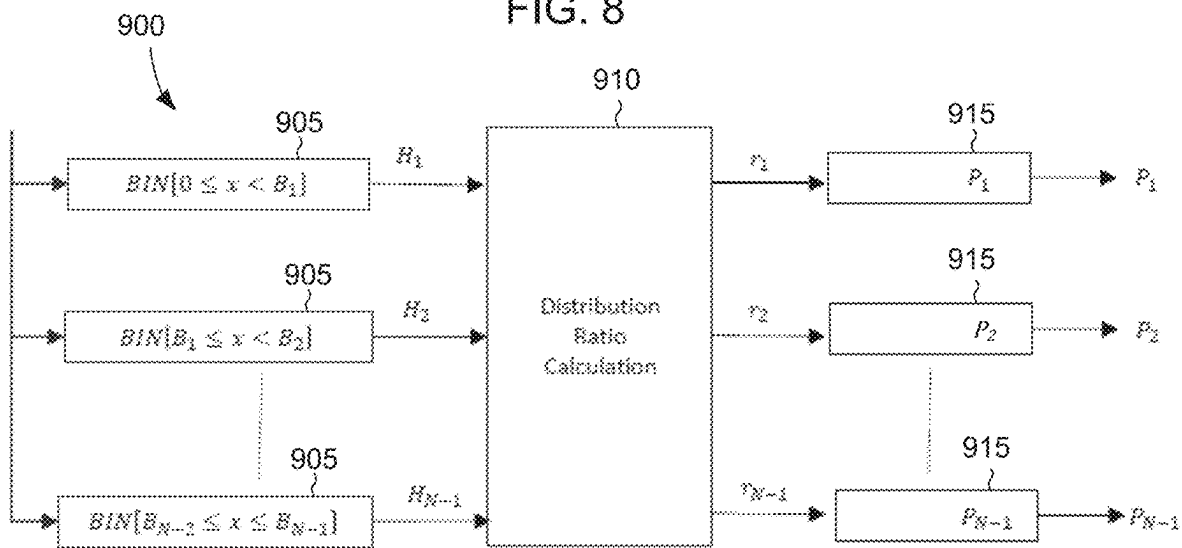


FIG. 9

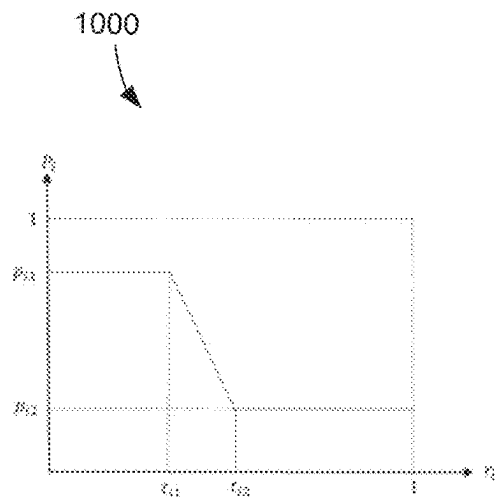


FIG. 10A

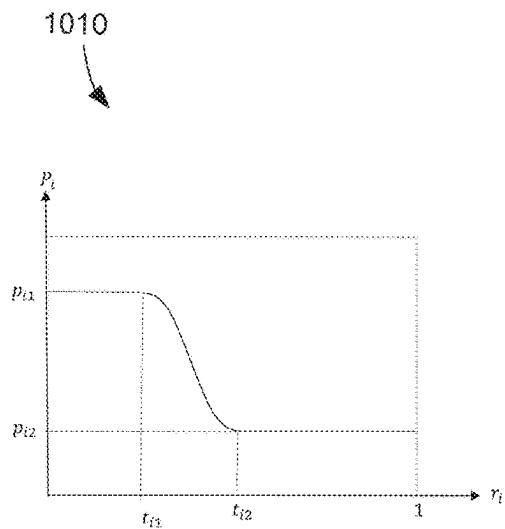


FIG. 10B

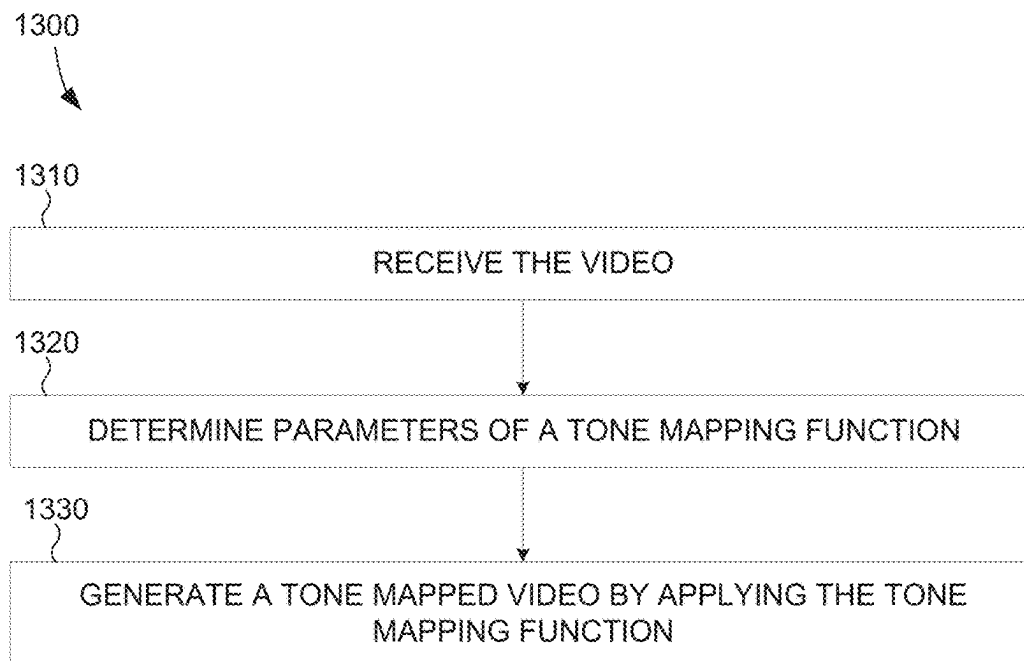


FIG. 13

1100

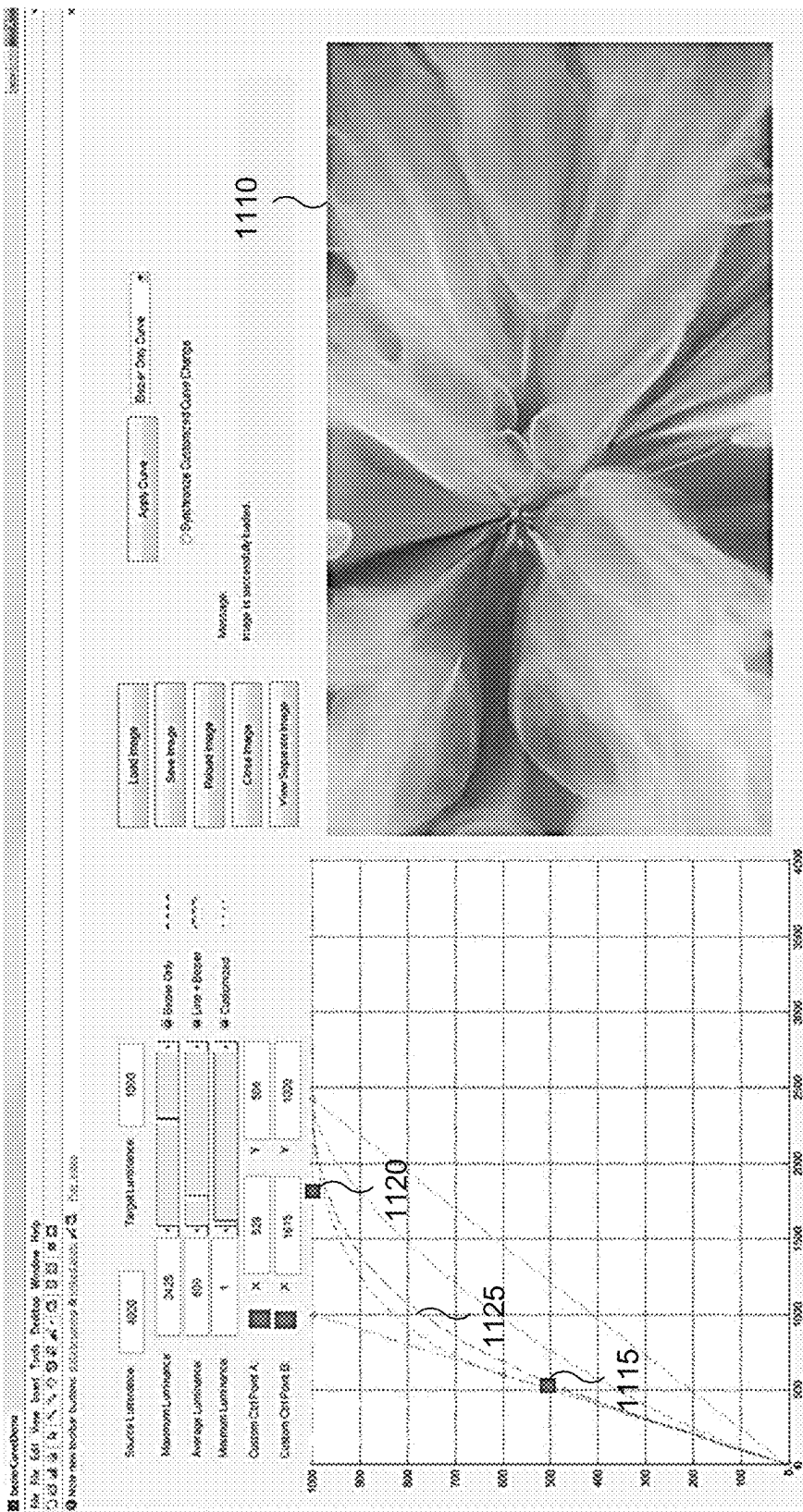


FIG. 11

1105

1

HDR TONE MAPPING METHODS WITH SCENE ADAPTIVE PARAMETERIZED TONE MAP FUNCTION

CROSS-REFERENCE TO RELATED APPLICATIONS AND CLAIM OF PRIORITY

This application claims priority under 35 U.S.C. § 120 as a continuation of U.S. patent application Ser. No. 17/512,603 filed on Oct. 27, 2021, which claims priority as a continuation of U.S. patent application Ser. No. 14/986,563 filed on Dec. 31, 2015 (now U.S. Pat. No. 11,176,646), which claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 62/184,667 filed on Jun. 25, 2015. All of these patent applications are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

This disclosure relates generally to video tone processing. More specifically, this disclosure relates to high dynamic range (HDR) tone mapping methods with parameterized tone map function.

BACKGROUND

At a studio, users can produce mastered videos using a mastering monitor to optimize or improve the luminance or brightness levels of the original videos. Often times, the mastering monitor used to master the video has different peak luminance or brightness levels than those of target displays upon which the videos are later displayed. Typically, the maximum luminance level of the mastering monitor is much larger than the maximum luminance level of the target display.

SUMMARY

This disclosure provides HDR tone mapping methods with parameterized tone map function.

In one embodiment, a computer-implemented method is provided. The method includes receiving, by a target display, a quantized video generated by a master display, the quantized video being mastered for a first range of luminance values associated with the master display. The method also includes receiving, by the target display, metadata from a tone mapping curve parameter extraction, the metadata comprising at least one maximum value of a luminance parameter of the quantized video as mastered for the first range of luminance values. The method also includes de-quantizing the quantized video to produce a de-quantized video. The method also includes applying, by at least one processor of the target display, a tone mapping function defined by a normalized Bezier curve to the de-quantized video. The method further includes generating, based on applying the tone mapping function by the at least one processor of the target display, a tone mapped video mastered for a second range of luminance values associated with the target display.

In another embodiment an apparatus is provided. The apparatus includes a display, a memory, and at least one processor operably connected to the memory. The at least one processor is configured to: receive a quantized video generated by a master display, the quantized video being mastered for a first range of luminance values associated with the master display; receive metadata from a tone mapping curve parameter extraction, the metadata compris-

2

ing at least one maximum value of a luminance parameter of the quantized video as mastered for the first range of luminance values; de-quantize the quantized video to produce a de-quantized video; apply a tone mapping function defined by a normalized Bezier curve to the de-quantized video; and generate a tone mapped video mastered for a second range of luminance values associated with the display.

In yet another embodiment, a non-transitory computer readable medium comprising program code is provided. The program code, when executed by at least one processor, causes the at least one processor to: receive a quantized video generated by a master display, the quantized video being mastered for a first range of luminance values associated with the master display; receive metadata from a tone mapping curve parameter extraction, the metadata comprising at least one maximum value of a luminance parameter of the quantized video as mastered for the first range of luminance values; de-quantize the quantized video to produce a de-quantized video; apply a tone mapping function defined by a normalized Bezier curve to the de-quantized video; and generate a tone mapped video mastered for a second range of luminance values associated with a display.

Other technical features may be readily apparent to one skilled in the art from the following figures, descriptions, and claims.

Before undertaking the DETAILED DESCRIPTION below, it may be advantageous to set forth definitions of certain words and phrases used throughout this patent document. The term “couple” and its derivatives refer to any direct or indirect communication between two or more elements, whether or not those elements are in physical contact with one another. The terms “transmit,” “receive,” and “communicate,” as well as derivatives thereof, encompass both direct and indirect communication. The terms “include” and “comprise,” as well as derivatives thereof, mean inclusion without limitation. The term “or” is inclusive, meaning and/or. The phrase “associated with,” as well as derivatives thereof, means to include, be included within, interconnect with, contain, be contained within, connect to or with, couple to or with, be communicable with, cooperate with, interleave, juxtapose, be proximate to, be bound to or with, have, have a property of, have a relationship to or with, or the like. The term “controller” means any device, system or part thereof that controls at least one operation. Such a controller may be implemented in hardware or a combination of hardware and software and/or firmware. The functionality associated with any particular controller may be centralized or distributed, whether locally or remotely. The phrase “at least one of,” when used with a list of items, means that different combinations of one or more of the listed items may be used, and only one item in the list may be needed. For example, “at least one of: A, B, and C” includes any of the following combinations: A, B, C, A and B, A and C, B and C, and A and B and C.

Moreover, various functions described below can be implemented or supported by one or more computer programs, each of which is formed from computer readable program code and embodied in a computer readable medium. The terms “application” and “program” refer to one or more computer programs, software components, sets of instructions, procedures, functions, objects, classes, instances, related data, or a portion thereof adapted for implementation in a suitable computer readable program code. The phrase “computer readable program code” includes any type of computer code, including source code, object code, and executable code. The phrase “computer

3

readable medium" includes any type of medium capable of being accessed by a computer, such as read only memory (ROM), random access memory (RAM), a hard disk drive, a compact disc (CD), a digital video disc (DVD), or any other type of memory. A "non-transitory" computer readable medium excludes wired, wireless, optical, or other communication links that transport transitory electrical or other signals. A non-transitory computer readable medium includes media where data can be permanently stored and media where data can be stored and later overwritten, such as a rewritable optical disc, an erasable memory device, static RAM, dynamic RAM, or flash memory.

Various functions described below can be implemented or supported by a processor coupled to a computer readable medium storing one or more computer programs. As such, the processor is a special purpose processor for performing the functions defined by the one or more computer programs.

Definitions for other certain words and phrases are provided throughout this patent document. Those of ordinary skill in the art should understand that in many if not most instances, such definitions apply to prior as well as future uses of such defined words and phrases.

BRIEF DESCRIPTION OF THE DRAWINGS

For a more complete understanding of this disclosure and its advantages, reference is now made to the following description, taken in conjunction with the accompanying drawings, in which:

FIG. 1 illustrates an example computing system in which various embodiments of the present disclosure may be implemented;

FIG. 2 illustrates a block diagram for generating a mastered video that can be remastered according to various embodiments of the present disclosure;

FIG. 3 illustrates a block diagram for a video display according to various embodiments of the present disclosure;

FIG. 4 illustrates a block diagram for generating a mastered video that can be remastered according to various embodiments of the present disclosure;

FIG. 5 illustrates a block diagram for a video display according to various embodiments of the present disclosure;

FIG. 6A illustrates an example graph for analyzing a video using a quantizer according to various embodiments of the present disclosure;

FIG. 6B illustrates an example graph for analyzing a video using a de-quantizer according to various embodiments of the present disclosure;

FIGS. 7A and 7B illustrate example display-side tone mapping according to the various embodiments of the present disclosure;

FIG. 8 illustrates an example parameterized tone mapping curve according to various embodiments of the present disclosure;

FIG. 9 illustrates a scene adaptive parameter extraction function according to the various embodiments of the present disclosure;

FIGS. 10A and 10B illustrate a function for calculating a control point from a distribution ratio according to the various embodiments of the present disclosure;

FIG. 11 illustrates an example user interface for tone mapping curve parameter extraction according to the various embodiments of the present disclosure;

FIG. 12 illustrates an example video processing device according to various embodiments of the present disclosure; and

4

FIG. 13 illustrates an example process for video tone mapping according to various embodiments of the present disclosure.

DETAILED DESCRIPTION

FIGS. 1 through 13, discussed below, and the various embodiments used to describe the principles of this disclosure in this patent document are by way of illustration only and should not be construed in any way to limit the scope of the disclosure. Those skilled in the art will understand that the principles of this disclosure may be implemented in any suitably arranged wireless communication system.

Embodiments of the present disclosure recognize that displaying the mastered video at on a display with a maximum luminance level that is lower than the maximum luminance level in the mastered video can reduce the quality of the video. For example, portions of the mastered video with luminance levels greater than the maximum luminance level of the target display may be clipped or limited to the maximum luminance level of the target display, thereby limiting the range of luminance values of the video displayed.

Accordingly, various embodiments of the present disclosure produce a tone mapped video using HDR tone mapping methods with scene adaptive parameterized tone map function. In one or more embodiments, the adaptive tone mapping based on local contrast may reduce costs while providing improved video quality regarding saturation of highlighted objects or scenes.

One or more embodiments relate to data for rendering images on a display device. In one embodiment, a method comprises one or more applications of tone mapping of images, graphics, and/or video (collectively referred to simply as "video" herein) to facilitate remastering content and/or to facilitate high dynamic range (HDR) displays. For example, one or more embodiments are described herein that can facilitate tone mapping for HDR displays using parameterized tone map functions. It will be appreciated that the various embodiments disclosed herein can be used either alone or in combination.

Taking an example where a studio already mastered a movie with a mastering monitor having maximum luminance level of M-nit, a user may want the mastered video to have a different peak luminance level of N-nit for a target display where $N < M$. In such case, the user may need to re-master the original video file using a mastering monitor having a maximum luminance level of N-nit. Embodiments of the present disclosure recognize that this process can be time intensive and costly. In another example, the user may use a tone mapping method using already mastered video to generate a tone mapped video with algorithms or software that down tone maps from the higher mastering monitor luminance level to the lower target display luminance level. Embodiments of the present disclosure recognize that this process can produce unsatisfactory results, such as, saturation of highlighted objects or scenes.

FIG. 1 illustrates an example computing system 100 in which various embodiments of the present disclosure may be implemented. The embodiment of the computing system 100 shown in FIG. 1 is for illustration only. Other embodiments of the computing system 100 could be used without departing from the scope of this disclosure.

As shown in FIG. 1, the system 100 includes a network 102, which facilitates communication between various components in the system 100 on one or more communication channels. The network 102 may communicate Internet Pro-

ocol (IP) packets, frame relay frames, or other information between network addresses. The network **102** may include one or more local area networks (LANs); metropolitan area networks (MANs); wide area networks (WANs); all or a portion of a global network, such as the Internet; or any other communication system or systems at one or more locations.

In various embodiments, network **102** includes a broadcast and broadband networks and communication channels for communicating video data (e.g., video files including metadata and audio content) to client devices **106-115**. The broadcasting elements of network **102**, such as cable and satellite communication links, provide broadcast of video data to client devices **106-115**. The broadband elements of network **102**, Internet, wireless, wireline, and fiber optic network links and devices, provide for streaming and download of video data.

The network **102** facilitates communications between one or more servers **104** and various client devices **106-115**. Each of the servers **104** includes any suitable computing or processing device that can provide computing services for one or more client devices. Each of the servers **104** could, for example, include one or more processing devices, one or more memories storing instructions and data, and one or more network interfaces facilitating communication over the network **102**. For example, one or more of the servers **104** may include processing circuitry for mastering a video or for tone mapping a video based on local contrast, as discussed in greater detail below.

Each client device **106-115** represents any suitable computing or processing device that interacts with at least one server or other computing device(s) over the network **102**. As will be discussed in greater detail below, the client devices **106-115** receive video (e.g., a mastered video file to be tone mapped or a tone mapped video file to be displayed) and may include processing circuitry for tone mapping a video based on local contrast. The client devices **106-115** each either includes or is connected to a display device for display of tone mapped video. In this example, the client devices **106-115** include a computer **106**, a mobile telephone or smartphone **108**, a personal digital assistant (PDA) **110**, a laptop computer **112**, tablet computer **114**; and a set-top box and/or television **115**. However, any other or additional client devices could be used in the communication system **100**. In this example, some client devices **108-114** communicate indirectly with the network **102**. For example, the client devices **108-110** communicate via one or more base stations **116**, such as cellular base stations or eNodeBs. Also, the client devices **112-115** communicate via one or more wireless access points **118**, such as IEEE 802.11 wireless access points. Each client device **112-115** can also receive or be used to create control points for a Bezier curve function. Note that these are for illustration only and that each client device could communicate directly with the network **102** or indirectly with the network **102** via any suitable intermediate device(s) or network(s).

Although FIG. **1** illustrates one example of a communication system **100**, various changes may be made to FIG. **1**. For example, the system **100** could include any number of each component in any suitable arrangement. In general, computing and communication systems come in a wide variety of configurations, and FIG. **1** does not limit the scope of this disclosure to any particular configuration. While FIG. **1** illustrates one operational environment in which various features disclosed in this patent document can be used, these features could be used in any other suitable system.

FIG. **2** illustrates a system **200** for generating a mastered video HDR tone mapping methods with a scene adaptive

parameterized tone map function according to embodiments of the present disclosure. The embodiment of the mastered video generating system **200** shown in FIG. **2** is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure.

During manual mastering block **205**, an input master video file F may be graded using a color grading module **210**. As outputs, the color grading module **210** provides a mastered video file f_M , which is fed as an input to a quantizer Q_n **215** and a metadata extraction module **220**. The quantizer Q_n **215** quantizes the mastered video file f_M to produce an n -bit quantized mastered video file $Q_n(f_M)$. The associated maximum luminance level of the video file f is L_{max} nit. The quantizer Q_n **215** quantizes f into 2^n levels (i.e., n -bit long binary code) with a pre-determined quantization curve.

The metadata extraction module **220** generates metadata about the mastered video file, for example, maximum luminance level of the mastering monitor **225** (M), maximum content luminance level (MaxCLL) or brightness, MaxFALL, or other properties about the mastered video file. The files f_M and $Q_n(f_M)$ as well as the metadata associated with the mastered video file are be stored in a storage device (e.g., such as storage device **915** in FIG. **9** discussed below) for transfer to and later display by a target display device (e.g., one of client devices **106-115**).

A mastering monitor **225** de-quantizes the quantized mastered video file for display of the video on the mastering monitor **225**. The de-quantization Q_n^{-1} is the inverse operation of the quantization operation of the quantizer Q_n **215**. By displaying the mastered video, the mastering monitor **225** provides visual feedback for continued or iterative adjustment of color tones for the manual mastering block **205**. However, the mastering monitor **225** may have very high contrast (e.g., has a maximum luminance level of M) in comparison with contrast available on a target display device (e.g., consumer-grade television sets, projectors, mobile phones, etc.) (e.g., having a maximum luminance level of N where $M > N$). As a result, the mastered video displayed on the mastering monitor **225** may look different than displayed on a target display device. For example, because the mastering monitor **225** can display brighter images than a typical home television set, the bright portions of the mastered video displayed on the home television set may clip and, thus, image details may be lost. For example, portions of in the mastered video having luminance levels above the maximum luminance level of the target display may all be displayed at the same luminance level (i.e., at the maximum luminance level of the target display).

FIG. **3** illustrates a block diagram for a video display **300** according to various embodiments of the present disclosure. The embodiment of the video display **300** shown in FIG. **3** is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure.

The example video display **300** includes a de-quantizer **305**, a parameterized tone mapping function **310**, a nit-to-bit function **315**, and an m -bit display **320**. The mastered video is received by the video display **300** and can be received with or separately from the metadata **330**. The mastered video is processed in the de-quantizer. The de-quantizer **415** de-quantizes an n -bit quantized mastered video file to produce the mastered video file.

Using the received metadata **330**, the parameterized tone mapping function **310** processes the mastered video. In certain embodiments, a Bezier curve function is used to process the mastered video. When a Bezier curve is used, control points are identified in the metadata. An amount of control points identified determines the order of the Bezier

curve function. The nit-to-bit function 315 decodes the mastered video. The video is then rendered for an m-bit display 320.

FIG. 4 illustrates a block diagram for generating a mastered video that can be remastered according to various embodiments of the present disclosure. The embodiment of the mastered video generating system 400 shown in FIG. 4 is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure.

During manual mastering block 405, an input master video file F may be graded using a color grading module 410. As outputs, the color grading module 410 provides a mastered video file f_M , which is fed as an input to a quantizer Q_n 415. The quantizer Q_n 415 quantizes the mastered video file f_M to produce an n-bit quantized mastered video file $Q_n(f_M)$. The associated maximum luminance level of the video file f is L_{max} nit. The quantizer Q_n 415 quantizes f into 4^n levels (i.e., n-bit long binary code) with a pre-determined quantization curve.

A mastering monitor 425 de-quantizes the quantized mastered video file for display of the video on the mastering monitor 425. The de-quantization Q_n^{-1} is the inverse operation of the quantization operation of the quantizer Q_n 415. By displaying the mastered video, the mastering monitor 425 provides visual feedback for continued or iterative adjustment of color tones for the manual mastering block 405. However, the mastering monitor 425 may have very high contrast (e.g., has a maximum luminance level of M) in comparison with contrast available on a target display device (e.g., consumer-grade television sets, projectors, mobile phones, etc.) (e.g., having a maximum luminance level of N where $M > N$). As a result, the mastered video displayed on the mastering monitor 425 may look different than displayed on a target display device. For example, because the mastering monitor 425 can display brighter images than a typical home television set, the bright portions of the mastered video displayed on the home television set may clip and, thus, image details may be lost. For example, portions of in the mastered video having luminance levels above the maximum luminance level of the target display may all be displayed at the same luminance level (i.e., at the maximum luminance level of the target display).

FIG. 5 illustrates a block diagram for a video display 500 according to various embodiments of the present disclosure. The embodiment of the video display 500 shown in FIG. 5 is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure.

The example video display 500 includes a de-quantizer 505, a parameterized tone mapping function 510, a nit-to-bit function 515, and an m-bit display 520. The mastered video is received by the video display 500. The mastered video is processed in the de-quantizer. The de-quantizer 415 de-quantizes an n-bit quantized mastered video file to produce the mastered video file.

The scene adaptive parameter extract function 530 generates metadata about the mastered video file, for example, maximum luminance level of a mastering monitor (M), maximum content luminance level (MaxCLL) or brightness, MaxFALL, or other properties about the mastered video file. The scene adaptive parameter extract function 530 determines the control points of a video for processing. The scene adaptive parameter extract function is described in greater detail along with FIG. 9.

Using the control parameters from the scene adaptive parameter extraction function 530, the parameterized tone mapping function 510 processes the mastered video. In certain embodiments, a Bezier curve function is used to

process the mastered video. An amount of control points identified determines the order of the Bezier curve function. The nit-to-bit function 515 decodes the mastered video. The video is then rendered for an m-bit display 520.

FIG. 6A illustrates an example graph 600 for analyzing a video using a quantizer according to various embodiments of the present disclosure. FIG. 6B illustrates an example graph 605 for analyzing a video using a de-quantizer according to various embodiments of the present disclosure.

The quantizer graph 600 can be used by the quantizer 215 or 415. The associate maximum luminance level off is L_{max} nit. Q_n quantifies f into 2^n levels (n-bit long binary code) with a pre-determine quantization curve.

The de-quantizer graph 605 can be used by the de-quantizer 305 or 505. The de-quantizer graph 605 is an inverse operation of Q_n . In ideal cases (L_{max} is a floating point), $x = Q_n^{-1}(Q_n(n)) = f$. But practically, there is a quantization error $|x - f|$ if L_{max} is an integer, as a function of n.

FIGS. 7A and 7B illustrate example display-side tone mapping according to the various embodiments of the present disclosure. FIG. 7A illustrates a video display similar to the video display 300 in FIG. 3. FIG. 7B illustrates a video display similar to the video display 500 in FIG. 5. In FIG. 7A the display 700 combines the parameterized tone mapping curve function 310 and the nit-to-bit function 315 into a combined tone mapping function 710. In FIG. 7B the display 705 combines the parameterized tone mapping curve function 510 and the nit-to-bit function 515 into a combined tone mapping function 715. The combined tone mapping functions 710 and 715.

FIG. 8 illustrates an example parameterized tone mapping curve 800 according to various embodiments of the present disclosure. FIG. 8 illustrates an example N-th order explicit Bezier curve 830 according to various embodiments of the present disclosure. In certain embodiments, an N-th order Bezier curve 830 is used for the parameterized tone mapping functions 310, 510, 710 and 715. In certain embodiments, an N-th order explicit Bezier tone mapping curve 830 can be determine using equation 1:

$$y = \sum_{k=0}^N \binom{N}{k} x^k (1-x)^{N-k} P_k,$$

where

$$\binom{N}{k} = \frac{N!}{k!(N-k)!}, \quad 0 \leq x, y \leq 1,$$

and where

$P = \{P_0, P_1, \dots, P_N\}$ are control points ($0 \leq P_i \leq 1$, $P_0 = 0$, $P_N = 1$). where N is the order of the Bezier function, x is an input value (e.g., normalized luminance value) for a pixel, P is a control point, and y is the output value (e.g., normalized luminance value) for the pixel.

FIG. 9 illustrates a scene adaptive parameter extraction function 900 according to the various embodiments of the present disclosure. The embodiment of the scene adaptive parameter extraction function 900 shown in FIG. 9 is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure.

The scene adaptive parameter extraction function 900 receives the mastered video file and distributes the video into bins 905 based on the luminance values of the one or more frames. The bins 905 are processed through a distribution ratio calculation function 910 to determine a com-

9

puted distribution of the luminance values, r_i . In certain embodiments, the r_i can be calculated using the equation:

$$r_i = \frac{H_i}{\sum_{k=1}^{N-1} H_k}$$

where r_i is the computed distribution of luminance value and H_i is the plurality of bins. The computed distribution of luminance values is used to calculate the control points for the Bezier function.

FIGS. 10A and 10B illustrate functions 1000 and 1010 for calculating a control point from a distribution ratio according to the various embodiments of the present disclosure. The embodiment of the function 1000 and 1010 shown in FIGS. 10A and 10B are for illustration only. Other embodiments could be used without departing from the scope of the present disclosure. Functions 1000 and 1010 are used to determine the control points, P_i , from the computed distribution of luminance value, r_i . The functions are for illustration only and any function could be used.

FIG. 11 illustrates an example user interface 1100 for tone mapping curve parameter extraction according to the various embodiments of the present disclosure. The embodiment of the user interface 1100 shown in FIG. 11 is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure. The user interface 1100 includes a control panel 1105 and a video panel 1110. The user interface 1100 shown below can be provided as a utility of the color grading module 210 of FIG. 2 or 410 of FIG. 4 for adjusting the tone map of the video. In certain embodiments, the effects of the selected tone mapping can be seen on the mastering monitor 205 and 405. The user interface 1100 allows the user to select the control points P1 and P2 (labeled as control points A 1115 and B 1120) as well other parameters.

After an image is loaded to the video panel 1110, control points A 1115 and B 1120 are automatically computed by image characteristics and display on the control panel 1105. Control points A 1115 and B 1120 can be manually adjusted by different user input methods, for example a mouse or touch screen. As the control points 1115 and 1120 are adjusted in the control panel 1105, the customized curve 1125 and the image in the video panel 1110 automatically adjust. The suitable curve can be selected and control points 1115 and 1120 are extracted.

FIG. 12 illustrates an example video processing device 1200 according to this disclosure. In particular, the video processing device 1200 illustrates example components that may be included in any one of the server 104 or the client devices 106-115 in FIG. 1 to implement one or more embodiments of the present disclosure.

As shown in FIG. 12, the video processing device 1200 includes a bus system 1205, which supports communication between at least one processor 1210, at least one storage device 1215, at least one communication interface 1220, at least one input/output (I/O) unit 1225, and a display 1240.

The processor 1210 executes instructions that may be loaded into a memory 1230. The processor 1210 may include any suitable number(s) and type(s) of processors or other devices in any suitable arrangement. Example types of processor 1210 include microprocessors, microcontrollers, digital signal processors, field programmable gate arrays, application specific integrated circuits, and discrete circuitry. For example, the processor 1210 may implement tone mapping operations performed by any of the processes and

10

systems 300-1100 being implemented in hardware or by executing stored instructions that causes the processor 1210 to perform the disclosed methods.

The memory 1230 and a persistent storage 1235 are examples of storage devices 1215, which represent any structure(s) capable of storing and facilitating retrieval of information (such as video data, program code, and/or other suitable information on a temporary or permanent basis). The memory 1230 may represent a random access memory or any other suitable volatile or non-volatile storage device(s). For example, the memory 1230 may contain instructions for implementing tone mapping techniques and/or may store streamed or buffered video data received from a server 104. The persistent storage 1235 may contain one or more components or devices supporting longer-term storage of data, such as a read-only memory, hard drive, Flash memory, or optical disc. For example, the persistent storage 1235 may contain video data such as a mastered video file and metadata associated with the mastered video file.

The communication interface 1220 supports communications with other systems or devices. For example, the communication interface 1220 could include a network interface card, a cable modem, a broadcast receiver, or a wireless transceiver facilitating communications over the network 102. The communication interface 1220 may support communications through any suitable physical or wireless communication link(s).

The I/O unit 1225 allows for input and output of data. For example, the I/O unit 1225 may provide a connection for user input through a keyboard, mouse, keypad, touchscreen, or other suitable input device. The I/O unit 1225 may also send output to the display 1240, printer, or other suitable output device.

The video processing device 1200 further includes or is connected to a display 1240, such as, for example, the target displays 345, 365, 510, 720, and 825. In one example embodiment, the video processing device 1200 may be video processing circuitry included in a set top box, cable box, computer, etc. connected to the display 1240 for tone mapped video to be displayed. In another example embodiment, the video processing device 1200 may be the set top box, cable box, computer, media player, etc. connected to the display 1240 for tone mapped video to be displayed. In yet another example embodiment, the video processing device 1200 may be a server connected to the target display 1240 over a network connection. In another example embodiment, the video processing device 1200 may include the display 1240 that the tone mapped video to be displayed on as well as the video processing circuitry for performing the tone mapping. For example, the video processing device 1200 may be a television, monitor, mobile phone, laptop computer, tablet computer, etc.

FIG. 13 illustrates an example process 1300 for video tone mapping in accordance with various embodiments of the present disclosure. For example, the process 1300 may be performed by the server 104 or the client devices 106-115 in FIG. 1. In particular, the process 1300 can be implemented by any one of the systems 300-500 and 700-705 in FIG. 3-5 or 7 discussed above, referred to collectively here as "the system." The embodiment of the process 1300 shown in FIG. 13 is for illustration only. Other embodiments could be used without departing from the scope of the present disclosure. While particular blocks of the process 1300 are described below as being performed by specific blocks of one of the systems 300-500 and 700-705, it will be understood that any suitable block can perform each of the blocks of the process 1300 in alternative example embodiments.

11

In block **1310**, the system receives the video for processing. For example, the video may be a mastered video file from a mastered video generating system **400**. The mastered video file can be color graded in a color grading module **410** and quantized by a quantizer **415**. The mastered video file can be tone mapped on a mastering monitor **425** with a high nit value, such as 4k nit.

In block **1320**, the system determines the parameters of a tone mapping function. For example, block **1320** may be implemented by scene adaptive parameter extract function **530** in FIG. 5. In certain embodiments, the parameters are determined using scene adaptive processing of one or more frames, such as a scene adaptive parameter extraction function **900**. The scene adaptive processing includes the system determining a distribution of luminance values of the one or more frames based on a plurality of bins, where each of the bins corresponds to a range of luminance values, and determining control points of the Bezier curve, such as Bezier curve **830**, as a function of the computed distribution of luminance values as discussed above with regard to FIG. 8. For example, the computed distribution of luminance values may be determined by the equation:

$$r_i = \frac{H_i}{\sum_{k=1}^{N-1} H_k},$$

where r is the computed distribution of luminance value and H is the plurality of bins. The system identifies an amount of control points in the parameters to determine an order of the Bezier function. the determined order Bezier function is expressed by:

$$y = \sum_{k=0}^N \binom{N}{k} x^k (1-x)^{N-k} P_k,$$

where N is the order of the Bezier function, x is an input value for a pixel, P is a control point, and y is the output value for the pixel.

In block **1330**, the system generates a tone mapped video by applying the tone mapped function. For example, the parameterized tone mapping function **510** processes the video using the Bezier curve **830**. The video is processed pixel-by-pixel to determine the tone mapped nit value for the video tone mapped for the video display **500**. The generated tone mapped video is then process using a nit-to-bit module **515** and rendered for the m-bit display **520**.

Although FIG. 13 illustrates an example of process for video tone mapping various changes could be made to FIG. 13. For example, while shown as a series of blocks, various blocks in each figure could overlap, occur in parallel, occur in a different order, or occur multiple times.

Although the present disclosure has been described with an exemplary embodiment, various changes and modifications may be suggested to one skilled in the art. It is intended that the present disclosure encompass such changes and modifications as fall within the scope of the appended claims.

None of the description in this application should be read as implying that any particular element, step, or function is an essential element that must be included in the claim scope. The scope of patented subject matter is defined only by the claims. Moreover, none of the claims is intended to

12

invoke 35 U.S.C. § 112(f) unless the exact words “means for” are followed by a participle.

What is claimed is:

1. A method comprising:

receiving a quantized video having a first maximum luminance level or brightness associated with a mastering display with which the quantized video was mastered;

de-quantizing the quantized video in order to generate a de-quantized video;

applying a scene-adaptive tone mapping function to the de-quantized video in order to generate a tone mapped video, wherein applying the scene-adaptive tone mapping function comprises:

determining a distribution of luminance values of one or more image frames based on multiple bins, each bin corresponding to a range of luminance values, the distribution comprising multiple distribution ratio values, each distribution ratio value associated with a different one of the bins and involving the luminance values within the associated bin; and

determining multiple control points of the scene-adaptive tone mapping function based on the distribution of luminance values, each control point associated with a different one of the distribution ratio values; and

displaying the tone mapped video on a target display; wherein the tone mapped video has a second maximum luminance level or brightness associated with the target display, the second maximum luminance level or brightness less than the first maximum luminance level or brightness.

2. The method of claim **1**, further comprising:

obtaining metadata associated with the quantized video, the metadata identifying the first maximum luminance level or brightness.

3. The method of claim **1**, wherein:

the scene-adaptive tone mapping function is associated with a Bezier curve; and

a number of the control points is associated with an order of the Bezier curve.

4. The method of claim **1**, wherein each of the distribution ratio values is determined by:

$$r_i = \frac{H_i}{\sum_{k=1}^{N-1} H_k},$$

where r represents the distribution of luminance values and H represents the bins.

5. The method of claim **1**, wherein the tone mapping function is expressed by:

$$y = \sum_{k=0}^N \binom{N}{k} x^k (1-x)^{N-k} P_k,$$

where N represents an order of a Bezier curve, x represents an input value for a pixel, P represents a control point, and y represents an output value for the pixel.

6. The method of claim **1**, further comprising:

displaying a user interface configured to receive at least one user input, the at least one user input defining the control points of the tone mapping function.

13

7. The method of claim 6, wherein:
 the at least one user input comprises coordinates to adjust
 a specified one of the control points of the tone map-
 ping function; and
 the method further comprises adjusting the tone mapping
 function based on the coordinates of the specified
 control point.
8. An apparatus comprising:
 at least one processing device configured to:
 receive a quantized video having a first maximum
 luminance level or brightness associated with a mas-
 tering display with which the quantized video was
 mastered;
 de-quantize the quantized video in order to generate a
 de-quantized video;
 apply a scene-adaptive tone mapping function to the
 de-quantized video in order to generate a tone
 mapped video, wherein, to apply the scene-adaptive
 tone mapping function, the at least one processing
 device is configured to:
 determine a distribution of luminance values of one
 or more image frames based on multiple bins,
 each bin corresponding to a range of luminance
 values, the distribution comprising multiple dis-
 tribution ratio values, each distribution ratio value
 associated with a different one of the bins and
 involving the luminance values within the associ-
 ated bin; and
 determine multiple control points of the scene-adap-
 tive tone mapping function based on the distribu-
 tion of luminance values, each control point asso-
 ciated with a different one of the distribution ratio
 values; and
 initiate display of the tone mapped video on a target
 display;
 wherein the tone mapped video has a second maximum
 luminance level or brightness associated with the target
 display, the second maximum luminance level or
 brightness less than the first maximum luminance level
 or brightness.
9. The apparatus of claim 8, wherein the at least one
 processing device is further configured to obtain metadata
 associated with the quantized video, the metadata identifi-
 ying the first maximum luminance level or brightness.
10. The apparatus of claim 8, wherein:
 the scene-adaptive tone mapping function is associated
 with a Bezier curve; and
 a number of the control points is associated with an order
 of the Bezier curve.
11. The apparatus of claim 8, wherein each of the distri-
 bution ratio values is determined by:

$$r_i = \frac{H_i}{\sum_{k=1}^{N-1} H_k},$$

where r represents the distribution of luminance values and
 H represents the bins.

12. The apparatus of claim 8, wherein the tone mapping
 function is expressed by:

$$y = \sum_{k=0}^N \binom{N}{k} x^k (1-x)^{N-k} P_k,$$

14

where N represents an order of a Bezier curve, x represents
 an input value for a pixel, P represents a control point, and
 y represents an output value for the pixel.

13. The apparatus of claim 8, wherein:
 the at least one processing device is further configured to
 initiate display of a user interface; and
 the user interface is configured to receive at least one user
 input defining the control points of the tone mapping
 function.

14. The apparatus of claim 13, wherein:
 the at least one user input comprises coordinates to adjust
 a specified one of the control points of the tone map-
 ping function; and
 the at least one processing device is further configured to
 adjust the tone mapping function based on the coordi-
 nates of the specified control point.

15. A non-transitory computer readable medium compris-
 ing program code that when executed causes at least one
 processor to:

- receive a quantized video having a first maximum lumi-
 nance level or brightness associated with a mastering
 display with which the quantized video was mastered;
 de-quantize the quantized video in order to generate a
 de-quantized video;
 apply a scene-adaptive tone mapping function to the
 de-quantized video in order to generate a tone mapped
 video, wherein the program code that when executed
 causes the at least one processor to apply the scene-
 adaptive tone mapping function comprises program
 code that when executed causes the at least one pro-
 cessor to:
 determine a distribution of luminance values of one or
 more image frames based on multiple bins, each bin
 corresponding to a range of luminance values, the
 distribution comprising multiple distribution ratio
 values, each distribution ratio value associated with
 a different one of the bins and involving the lumi-
 nance values within the associated bin; and
 determine multiple control points of the scene-adaptive
 tone mapping function based on the distribution of
 luminance values, each control point associated with
 a different one of the distribution ratio values; and
 initiate display of the tone mapped video on a target
 display;
 wherein the tone mapped video has a second maximum
 luminance level or brightness associated with the target
 display, the second maximum luminance level or
 brightness less than the first maximum luminance level
 or brightness.

16. The non-transitory computer readable medium of
 claim 15, further comprising program code that when
 executed causes the at least one processor to obtain metadata
 associated with the quantized video, the metadata identifi-
 ying the first maximum luminance level or brightness.

17. The non-transitory computer readable medium of
 claim 15, wherein:

- the scene-adaptive tone mapping function is associated
 with a Bezier curve; and
 a number of the control points is associated with an order
 of the Bezier curve.

18. The non-transitory computer readable medium of
 claim 15, wherein each of the distribution ratio values is
 determined by:

$$r_i = \frac{H_i}{\sum_{k=1}^{N-1} H_k},$$

15

where r represents the distribution of luminance values and H represents the bins.

19. The non-transitory computer readable medium of claim **15**, wherein the tone mapping function is expressed by:

5

$$y = \sum_{k=0}^N \binom{N}{k} x^k (1-x)^{N-k} P_k,$$

10

where N represents an order of a Bezier curve, x represents an input value for a pixel, P represents a control point, and y represents an output value for the pixel.

20. The non-transitory computer readable medium of claim **15**, further comprising program code that when executed causes the at least one processor to:

initiate display of a user interface that is configured to receive at least one user input, the at least one user input comprising coordinates to adjust a specified one of the control points of the tone mapping function; and adjust the tone mapping function based on the coordinates of the specified control point.

* * * * *

16